

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations**

January 2023

19MEM102 – INTRODUCTION TO SELF DRIVING CARS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain environment perception module of the software architecture for an autonomous vehicle.	10	L*2	1	1
	b) What are proprioceptive sensors? Explain. Draw a sketch to emphasize the different perception sensors mounted on a car.	10	L1	2	1
2.	a) How to choose a hardware configuration by performing sensor coverage analysis for both longitudinal and transverse cases for highway and city driving.	10	L1	1	1
	b) Discuss the following with an example "Making driving decisions falls under the bigger umbrella of planning"	10	L2	2	2
3.	a) What is the main classification of sensors? Explain the characteristics of Camera and LIDAR.	10	L2	1	1
	b) Explain the localization map and occupancy grid map in autonomous driving.	10	L2	2	1
Unit – II					
4.	a) What is the safety Ideology followed by General Motors? Explain.	10	L2	3	1
	b) Explain basics of 2D kinematics using simple robot motion kinematics	10	L2	4	1
5.	a) Explain the five level safety approach designed by Waymo.	10	L2	3	1
	b) Explain typical powertrain model of a car with a block diagram. Write the dynamic equations of each powertrain components.	10	L2	4	1
6.	a) Explain probability fault tree analysis. How Failure Modes and Effects Analysis (FMEA) is different?	10	L2	3	1
	b) What are the different steps to build a typical dynamic model for a rigid body? Formulate dynamic model for a translational system with an example.	10	L1	4	1
Unit – III					
7.	a) What is the concept of receding horizon control? Draw MPC structure using a block diagram.	10	L1	5	1
	b) What is the basic concept used to control the path in lateral vehicle control? Explain	10	L1	5	1
8.	a) What is longitudinal cruise control? Explain with a block diagram.	10	L2	5	1
	b) Explain the architecture of vehicle control strategy with a block diagram.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
